ModelIt!

Teacher's Guide

Will Brain-Computer Interfaces Change Humanity?

Modeling Neural Signal Processing, Brain Plasticity, and Human-Machine Integration

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-LS1-2, HS-LS1-3, HS-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS1-2, HS-LS1-3, HS-ETS1-1

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask questions about how neural signal quality, decoding algorithms, and tissue response interact to determine BCI performance, and define the engineering problems that must be solved for clinical viability.
Disciplinary Core Idea	**LS1.A Structure and Function / LS1.D Information Processing** — Students model how the hierarchical organization of neural systems — from individual neurons to population codes to cortical networks — determines the signals available for BCI decoding, and how feedback mechanisms enable brain-machine co-adaptation.
Crosscutting Concept	**Cause and Effect / Structure and Function** — Students trace the causal chain from electrode design through signal quality, decoding accuracy, and ultimately functional capability, understanding how the physical structure of the interface determines its functional performance.

CAST Testing Alignment

MULTIPLE CHOICE:

An intracortical electrode array records from 200 neurons in motor cortex with a signal-to-noise ratio of 5:1. After 18 months of glial scarring, the tissue response reduces the signal-to-noise ratio to 2:1 and the number of detectable neurons to 80. If decoding accuracy scales approximately with the square root of the number of clearly recorded neurons, what is the approximate percentage decrease in decoding accuracy, and what compensatory strategies could maintain function?

CONSTRUCTED RESPONSE:

Using evidence from your model, argue whether brain-computer interfaces should be developed only for medical applications (restoring function to people with disabilities) or should also be pursued for human augmentation (enhancing normal cognitive capabilities). Address the signal quality requirements, surgical risks, long-term biocompatibility, ethical considerations, and societal implications in your response.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human brain is the most complex object in the known universe — 86 billion neurons forming 100 trillion synaptic connections, processing information through electrochemical signals that produce every thought, movement, sensation, and emotion you experience. For centuries, this organ was a black box: we could observe behavior but not the neural activity producing it. Brain-computer interfaces represent the first technology capable of directly reading — and potentially writing — the brain's electrical language, creating a communication channel between biological neural networks and digital computing systems. The implications range from the profoundly medical (restoring movement to paralyzed patients, giving voice to those who cannot speak) to the philosophically challenging (merging human cognition with artificial intelligence, raising questions about identity, privacy, and what it means to be human).

How Brain-Computer Interfaces Work

BCIs detect the electrical signals produced by neural activity and translate them into digital commands. The brain's electrical signals exist at multiple scales: individual action potentials (spikes from single neurons, lasting 1-2 milliseconds), local field potentials (collective activity from thousands of neurons in a small region), electrocorticographic signals (activity recorded from the brain's surface), and electroencephalographic signals (activity recorded through the skull from the scalp). Invasive BCIs like the Utah array — a 4x4 millimeter grid of 96 silicon microelectrodes that penetrate 1.5 millimeters into the cortex — record individual neuron spikes with high resolution but require brain surgery. Electrocorticographic BCIs place electrode strips on the brain surface, providing moderate resolution with less tissue penetration. Noninvasive EEG records through the scalp, requiring no surgery but receiving signals that have been blurred and attenuated by the skull. The detected signals are amplified, filtered, digitized, and fed into machine learning algorithms that learn the correlation between neural patterns and intended actions.

Neural Decoding and Machine Learning

The core challenge of BCI technology is neural decoding: determining what a person intends to do from the electrical activity of their brain. Motor BCIs decode intended movements by recording from motor cortex — the brain region that plans and executes physical actions. When a paralyzed person imagines moving their arm, motor cortex neurons fire in patterns similar to actual movement, and these patterns can be decoded into robotic arm commands. Speech BCIs decode intended speech by recording from speech motor cortex — the regions that control the tongue, lips, jaw, and larynx. Recent breakthroughs at Stanford and UC San Francisco have achieved speech decoding rates of 60-90 words per minute by decoding attempted speech movements from paralyzed patients. The machine learning algorithms — typically recurrent neural networks or Kalman filters — are trained on labeled data where the patient attempts known movements or words while neural activity is recorded. Over days to weeks of training, the decoder learns to map neural patterns to intended outputs with increasing accuracy.

The Tissue Response Challenge

The brain treats any implanted device as a foreign invader. Within hours of implantation, microglia (the brain's immune cells) surround the electrodes and begin an inflammatory response. Over weeks to months, astrocytes form a dense glial scar around each electrode, increasing the electrical impedance between the electrode and nearby neurons. This scar tissue progressively degrades signal quality: neurons within the scar zone die or migrate away, and the increased impedance reduces the amplitude of detectable signals. Studies show that intracortical arrays lose 50-80% of their initially recorded neurons within the first year. Some arrays continue functioning for years by recording from a changing population of nearby neurons, but signal quality generally declines. Addressing this tissue response is the primary challenge for long-term BCI viability. Approaches include smaller and softer electrode materials that match brain tissue mechanical properties, anti-inflammatory coatings, wireless systems that eliminate the infection risk of percutaneous connectors, and novel materials like carbon fiber electrodes that provoke less immune response.

Current Achievements and the Augmentation Frontier

The medical achievements of BCIs are remarkable. The BrainGate clinical trial has enabled paralyzed participants to control computer cursors, robotic arms, and powered wheelchairs using thought. Neuralink's PRIME study implanted its N1 device in a

quadriplegic patient in 2024, enabling cursor control and gaming. Synchron's Stentrode — implanted through a blood vessel without open brain surgery — has enabled ALS patients to text and browse the internet. Stanford researchers achieved 90+ words per minute speech decoding from a paralyzed patient. These achievements demonstrate that BCIs can restore meaningful function. The augmentation frontier — enhancing normal human capabilities rather than restoring lost ones — remains largely theoretical. Augmentation would require dramatically higher information transfer rates, bidirectional read-write capability, long-term biocompatibility measured in decades, and answers to profound ethical questions about mental privacy, cognitive liberty, and equitable access. The gap between current medical BCIs (restoring basic function to people with severe disabilities) and the augmentation vision (merging human and machine intelligence) is enormous in both technology and ethics.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven key components of a brain-computer interface system: electrode count and density, signal-to-noise ratio, decoding algorithm accuracy, tissue response, interface invasiveness, brain adaptation rate, and information transfer rate.

E **Establish Relationships**

Students map relationships showing that electrode count and interface invasiveness are the primary design choices determining signal-to-noise ratio, which feeds into decoding accuracy, while tissue response progressively degrades signal quality over time, and brain adaptation provides a biological compensation mechanism.

V **Visualize & Model**

Students build a computational model showing the seven-component BCI system with information transfer rate as the key performance outcome and tissue response as the primary degradation factor.

E **Evaluate Outcomes**

Students run motor prosthesis, noninvasive EEG, and long-term degradation scenarios to discover the performance envelope of current BCI technology and the biological barriers that constrain it.

R **Revise & Extend**

Students add bidirectional stimulation, wireless data transmission, or neural privacy to model more advanced and more ethically complex BCI applications.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"In 2024, a man paralyzed from the neck down played video games, browsed the internet, and moved a robotic arm — all using nothing but his thoughts. A chip in his brain read his neural signals and translated intention into action. This is not science fiction. It is happening right now. The question is: where does it go next?"*

Watch For

Open with video or images from BrainGate or Neuralink trials showing paralyzed patients controlling devices with thought. Ask: would you get a brain implant if it let you control any device with your mind? What if it could also read your thoughts? Catalog reactions.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the technology that bridges the gap between biological brain and digital machine. You will discover how it works, why it degrades over time, and why the hardest questions are not about engineering but about what we want technology to do to the human mind."*

Watch For

Have students read objectives. Pre-teach 'neural signal' by showing actual electrode recordings from published research. Pre-teach 'neuroplasticity' as the brain's ability to rewire itself — this is what makes BCIs possible.

Slide 3: Big Question

3 min

Say: *"We CAN — we already have. Paralyzed people are controlling devices with thought right now. The question is not whether it works for medical applications but whether we should extend it to healthy humans. Should you be able to think a text message instead of typing it? Should your employer be able to monitor your brain activity? Should your memories be downloadable?"*

Watch For

Quick-write: Students write three things they would want a brain-computer interface to do and three things they would never want it to do. Share out and discuss why the boundary between exciting and terrifying is so personal.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where electrical engineering meets neuroscience meets ethics — where the technical question of how many neurons you can record from leads directly to the philosophical question of how much of*

human thought should be accessible to machines."

Watch For

Preview LEVER steps. Emphasize that this model uniquely combines engineering performance with biological degradation and ethical implications — all three are essential.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components. Two you choose: how many electrodes and how invasive the surgery. Five that biology and physics determine. The most surprising insight: the brain does not just passively produce signals — it actively learns to communicate with the machine."*

Watch For

Guide through components. Show the Utah array at actual size (4x4 mm). Compare electrode count (96-1,024) to total neuron count (86 billion). Discuss why brain adaptation is not just helpful but essential.

Slide 6: Activity 2: Connections

8 min

Say: *"Everything flows through signal quality. Better electrodes give better signals. Better signals give better decoding. Better decoding gives higher information transfer. But the brain fights back — the tissue response degrades signals over time, and the brain adaptation tries to compensate. It is a three-way dynamic between engineering, immunology, and neuroplasticity."*

Watch For

Help students map the signal quality chain from electrodes through decoding to output. Highlight the tissue response as the primary threat. Calculate: if signal degrades 50% per year but brain adaptation compensates 30%, what is net performance after two years?

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios showing the current state of the art, the limitations of noninvasive approaches, and the biological clock that limits every implant."*

Watch For

Students predict performance before each scenario. The motor cortex scenario shows impressive capability. The EEG scenario shows the signal quality penalty of noninvasive approaches. The degradation scenario reveals the ticking clock of tissue response. Discuss what this means for patients relying on implants.

Slide 8: Discoveries

5 min

Say: *"BCIs are a genuine medical breakthrough for paralysis and communication disorders. The leap from restoration to augmentation — from helping disabled patients to enhancing healthy humans — is not just a bigger version of the same problem. It requires solving the tissue response, achieving dramatically higher data rates, and answering ethical questions that humanity has never faced before."*

Watch For

Lead discussion on the restoration-augmentation distinction. Most students can agree that restoring movement to a paralyzed person is ethical. Where does the ethical consensus break? Enhancing memory? Reading emotions? Thought surveillance? Map the class's ethical boundary.

Slide 9: STEM Challenge

12 min

Say: *"A neuroengineering company needs your BCI design for a specific patient population. The technology must work, the biology must cooperate, and the ethics must be addressed."*

Watch For

Groups choose between motor prosthesis, speech synthesis, or seizure intervention applications and design complete BCI systems. Must specify electrode technology, placement, decoding approach, biocompatibility strategy, and ethical informed consent framework. Present with model evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Electrode Count and Density, Interface Invasiveness
- **INTERNAL:** Signal-to-Noise Ratio, Decoding Algorithm Accuracy, Tissue Response, Brain Adaptation Rate, Information Transfer Rate

Relationships

- **Electrode Count and Density to Signal-to-Noise Ratio:** POSITIVE (+) with diminishing returns — More electrodes sampling more neurons provide richer signal data with more redundancy, improving the overall signal-to-noise ratio. However, returns diminish because additional electrodes in the same brain region record overlapping neural populations, and very high electrode density can increase tissue damage and inflammatory response.
- **Tissue Response to Signal-to-Noise Ratio:** NEGATIVE (-) — Progressive glial scarring around implanted electrodes increases electrical impedance and pushes neurons away from electrode tips, directly reducing the amplitude of recorded neural signals relative to background noise. This degradation is the primary factor limiting long-term BCI performance.
- **Signal-to-Noise Ratio to Decoding Algorithm Accuracy:** POSITIVE (+) — Higher signal quality provides the machine learning decoder with cleaner, more discriminable neural patterns to classify, directly improving the accuracy of decoded intentions. Below a critical signal-to-noise threshold, decoding accuracy drops sharply as neural patterns become indistinguishable from noise.
- **Brain Adaptation Rate to Decoding Algorithm Accuracy:** POSITIVE (+) — Faster and more complete brain adaptation means the user learns to produce more consistent, distinctive neural patterns that the decoder can more easily classify. This co-adaptation between brain and machine is essential — the brain effectively learns to speak the machine's language while the machine learns to understand the brain's language.
- **Interface Invasiveness to Signal-to-Noise Ratio:** POSITIVE (+) — More invasive interfaces — electrodes placed closer to or within neural tissue — record stronger, less attenuated signals with higher spatial resolution. Scalp EEG signals are 100-1,000 times weaker than intracortical recordings because the skull and scalp blur and attenuate the electrical fields.

Key Discoveries

- The number of neurons recorded is less important than the quality and diversity of neural signals — a well-placed array recording clearly from 100 neurons in motor cortex can decode arm movements more accurately than a poorly placed array recording noisy signals from 1,000 neurons, because movement is encoded in specific neural population patterns
- Brain plasticity is not just a bonus but a requirement for BCI function — the brain actively learns to produce neural patterns that the decoder can interpret, and this co-adaptation between brain and machine is what enables seemingly impossible performance from a tiny fraction of recorded neurons
- The tissue response to implanted electrodes is the primary factor limiting long-term BCI performance — glial scarring progressively isolates electrodes from neurons, degrading signal quality over months to years, creating a race between adaptive algorithms and biological encapsulation
- The gap between current BCI capability and science fiction augmentation is enormous — even the best current systems achieve information transfer rates far below natural communication, and the leap from restoring lost function to enhancing normal function requires orders-of-magnitude improvements in every system parameter

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Brain-computer interfaces can read your thoughts like reading a book"	Current BCIs decode motor intentions (what you want your body to do) from specific brain regions with limited resolution — they cannot read thoughts, memories, emotions, or inner speech in any general sense. The decoding works because motor cortex produces relatively organized, repeatable patterns when a person attempts specific movements. Higher cognitive functions like abstract thought, memory recall, and emotional processing involve distributed activity across billions of neurons in many brain regions simultaneously, producing patterns far too complex for current electrode arrays to capture or algorithms to decode. The model shows that information transfer rates of current BCIs are orders of magnitude below the bandwidth that would be required for thought reading. Reading specific motor intentions from motor cortex is possible; reading the content of your thoughts from the whole brain is not.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Brain implants last forever once installed"	Brain implants face progressive biological degradation that limits their functional lifetime. The brain's immune system treats the implant as a foreign body, triggering an inflammatory response that builds glial scar tissue around each electrode over months. This scar tissue increases electrical impedance (making signals weaker), pushes neurons away from electrode tips (reducing the number of recordable cells), and can physically damage both the tissue and the device. Studies show 30-80% signal degradation in the first year for intracortical arrays. Current devices may maintain clinically useful function for 3-5 years with adaptive algorithms, but no current technology has demonstrated stable performance over a full human lifespan. The model quantitatively tracks this degradation and shows why biocompatibility is the primary engineering challenge for long-term BCIs.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Noninvasive BCIs will soon match the performance of brain implants"	The fundamental physics of electrical signal propagation places hard limits on noninvasive BCI performance. Neural signals generated in the cortex must pass through cerebrospinal fluid, the skull (a poor electrical conductor), and the scalp before reaching surface electrodes. This journey attenuates signals by 100-1,000 times and blurs spatial resolution from 100 micrometers (intracortical) to 2-3 centimeters (scalp EEG), mixing signals from millions of neurons into an indistinguishable average. No amount of computational processing can recover spatial information that was lost during transmission through the skull. The model shows that this signal quality gap directly translates into an information transfer rate gap: 30-50 bits per second for invasive versus 1-5 bits per second for noninvasive. Noninvasive BCIs can improve through

	better algorithms and higher-density electrode arrays, but they cannot close this fundamental gap without either making the recording invasive or finding a completely new signal modality.